

Unit 5

Name	Gender	Fever	Cough	Test-1	Test-2	Test-3	Test-4
Jack	M	Y	N	P	N	N	A
Mary	F	Y	N	P	A	P	N
Jim	M	Y	P	N	N	N	A

Gender is a symmetric variable. P & Y = 1 and N & A = 0.
So, rewriting the table above:

Name	Gender	Fever	Cough	Test-1	Test-2	Test-3	Test-4
Jack	M	1	0	1	0	0	0
Mary	F	1	0	1	0	1	0
Jim	M	1	1	0	0	0	0

Jaccard's coefficient:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

$$J(\text{Jack}, \text{Mary}) = (1+1)/(1+1+1) = 0.67$$

$$J(\text{Jack}, \text{Jim}) = 1/(1+1+1) = 0.33$$

$$J(\text{Jim}, \text{Mary}) = 1/(1+1+1+1) = 0.25$$